



POZNAN UNIVERSITY OF TECHNOLOGY

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Master's thesis

**STEGANOGRAPHIC WATERMARKING FOR DATA POISONING
IN GENERATIVE IMAGE MODELS**

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Abstract

Abstract

The rapid growth of generative image models has raised concerns about the unauthorised use of artists' work for model training. This thesis investigates steganographic watermarking as a data poisoning mechanism in style-specific fine-tuning of latent diffusion models.

Experiments were conducted using Stable Diffusion v1.5, fine-tuned via Low-Rank Adaptation on a dataset of 559 Van Gogh paintings. Four watermarking strategies were evaluated: a classical frequency-domain method (DWT-DCT), a visible geometric marker, a colour channel shift, and an adversarial perturbation computed against the model's VAE encoder. Each method was assessed for watermark invisibility on training images and model deterioration in generated outputs.

Results show that classical frequency-domain watermarks do not survive VAE encoding and produce no measurable effect on the trained model. Colour channel shifts survive but are visibly detectable and cause distribution shifts rather than genuine deterioration. The adversarial perturbation method is the only approach that approaches the combination of partial invisibility and meaningful poisoning. A counterintuitive finding is that poisoning 50% of the training data produces stronger model deterioration than poisoning the full dataset.

The source code for all experiments is available at: <https://github.com/filo12345678910/steganography>.

Streszczenie

Dynamiczny rozwój modeli generatywnych obrazów wywołał poważne obawy dotyczące nieautoryzowanego wykorzystania twórczości artystów do trenowania modeli. Niniejsza praca bada steganograficzny watermarking jako mechanizm zatrutowania danych w kontekście dostrajania modeli dyfuzji do konkretnego stylu artystycznego.

Eksperymenty przeprowadzono na modelu Stable Diffusion v1.5, dostrojonym metodą Low-Rank Adaptation na zbiorze 559 obrazów Van Gogha. Przetestowano cztery strategie watermarkingu: klasyczną metodę częstotliwościową DWT-DCT, widoczny znacznik geometryczny, przesunięcie kanału kolorów oraz perturbację adversaryjną obliczaną bezpośrednio przeciwko enkoderowi VAE modelu. Każdą metodę oceniano pod kątem niewidoczności watermarku w danych treningowych oraz pogorszenia jakości generowanych obrazów.

Wyniki pokazują, że klasyczne watermarki częstotliwościowe nie przeżywają kodowania VAE i nie mają mierzalnego wpływu na wytrenowany model. Przesunięcia kanałów kolorów przeżywają kodowanie, ale są widoczne i powodują jedynie zmianę rozkładu generowanych obrazów. Metoda perturbacji adversaryjnej jako jedyna zbliża się do połączenia częściowej niewidoczności i skutecznego zatrutowania. Nieoczekiwanym wynikiem jest to, że zatrucie 50% danych treningowych powoduje silniejsze pogorszenie modelu niż zatrucie całego zbioru.

Kod źródłowy wszystkich eksperymentów dostępny jest pod adresem: <https://github.com/filo12345678910/steganography>.

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Chapter 1

Introduction

1.1 Context

The rapid progress of generative artificial intelligence has significantly transformed the creation and spread of digital visual content. Text-to-image models based on diffusion architectures have enabled the synthesis of high-quality images at scale, often trained on large collections of publicly available or scraped artworks. This development has raised substantial concerns related to copyright infringement, unauthorised style imitation, and the misuse of artists' works without consent.

In parallel, steganography and digital watermarking have evolved from classical signal-processing techniques into learning-based methods capable of embedding information in a manner that is difficult to detect yet robust to transformations. Traditionally, these techniques have been used for copyright protection, ownership verification, and covert communication. More recently, their adversarial potential has gained attention, particularly in the context of machine learning systems trained on large-scale data.

An emerging line of research explores the idea of intentionally modifying training data to influence or degrade the behaviour of machine learning models, a phenomenon commonly referred to as data poisoning. When applied to generative models, such attacks can aim not only to reduce output quality but also to induce systematic artefacts, embed detectable signals, or disrupt the learning of specific visual styles. Combining steganographic watermarking with data poisoning introduces a novel defensive perspective, where content creators may embed imperceptible signals into artworks to protect against unauthorised model training.

1.2 Goal

The goal of this thesis is to investigate steganographic watermarking as a mechanism for data poisoning in text-to-image generative models, with a focus on style-specific fine-tuning scenarios. The work aims to analyse whether imperceptibly watermarked images can be used to influence re-trained diffusion models, either by degrading generation quality, introducing identifiable artefacts, or disrupting the learning of specific visual styles.

The investigation is conducted using Stable Diffusion v1.5 as the base generative model, fine-tuned via Low-Rank Adaptation on a dataset of Van Gogh paintings. Van Gogh's distinctive and recognisable style provides a well-defined target for style transfer and a clear visual reference for measuring deterioration. Four watermarking strategies of varying aggressiveness are evaluated - a classical frequency-domain method, a visible geometric marker, a color channel shift, and an

adversarial perturbation computed against the model’s VAE encoder - each applied to the training dataset and assessed for both imperceptibility and poisoning effectiveness.

The experiments reveal that classical frequency-domain watermarks do not survive the VAE encoding step and produce no measurable effect on the trained model, while adversarial perturbation is the only method that approaches the combination of partial invisibility and meaningful poisoning required for a practical steganographic attack. The trade-off between imperceptibility and poisoning effectiveness is a central theme throughout the work.

1.3 Research Questions

The objectives of this thesis are formalised through the following research questions.

RQ1: How can steganographic and watermarking techniques be applied to visual datasets used for training or fine-tuning generative image models?

RQ2: To what extent do embedded steganographic signals survive the training process of diffusion-based text-to-image models?

RQ3: Can steganographically watermarked artworks be used to poison style-specific fine-tuning, resulting in degraded or altered generative outputs?

RQ4: What trade-offs arise between imperceptibility, survival through latent encoding, and poisoning effectiveness?

1.4 Thesis Structure

The remainder of this thesis is organised as follows. Chapter 2 reviews the relevant literature on data poisoning, backdoor attacks in generative models, steganography and watermarking, and the Parasite framework. Chapter 3 provides the theoretical background covering diffusion models, latent diffusion and the VAE, Low-Rank Adaptation, and DWT-DCT watermarking. Chapter 4 describes the experimental infrastructure, covering the dataset preparation, base model selection, and the fine-tuning approach adopted throughout the thesis. Chapter 5 details the iterative process of training and selecting the baseline model, including initial experiments, the discovery of the correct training strategy, the hyperparameter search, and the final experiment setup used for all poisoning runs. Chapter 6 presents the watermarking and poisoning experiments, evaluates each method using a two-group scoring framework, and concludes with a comparative analysis across all methods. Chapter 7 concludes the thesis by summarising the findings, answering the research questions, discussing limitations, and identifying directions for future work.

Chapter 2

Literature Review

This chapter reviews prior work relevant to steganographic watermarking and data poisoning in generative image models. The discussion progresses from general concepts in data poisoning and information hiding to recent attacks on diffusion models, culminating in a focused analysis of the Parasite framework. The selection of literature and organization of this review follows the systematic literature review (SLR) protocol described in Appendix A.

2.1 Data Poisoning in Machine Learning

Data poisoning refers to attacks in which an adversary manipulates the training dataset to influence a model’s learned behavior. Prior research shows that modern machine learning models can be highly sensitive to carefully crafted training samples, even when the overall data distribution appears unchanged.

Early work by Biggio et al. [3] formalized data poisoning as an optimization problem, demonstrating that modifying a limited subset of training data can significantly degrade model performance or steer decision boundaries in attacker-controlled directions. This work established that an attacker does not need full control over the training set, but can exploit the learning process itself by injecting strategically designed samples.

Subsequent studies extended these ideas to deep neural networks and showed that poisoning attacks can induce targeted behaviors rather than merely reducing overall accuracy. In particular, Shafahi et al. [14] demonstrated that clean-label poisoning—where all labels remain correct—can still cause specific target inputs to be misclassified after training, while preserving high performance on benign data.

Importantly, this work also revealed that only a very small number of poisoned samples may be sufficient to influence model behavior, especially in retraining or transfer learning scenarios. Fine-tuned models inherit representations from pretrained networks and can therefore over-weight individual samples in small retraining datasets. This effect is particularly relevant for settings where models are adapted using limited data, such as artist- or style-specific fine-tuning, where poisoned samples can exert disproportionate influence and enable subtle, hard-to-detect manipulations.

2.2 Backdoor and Poisoning Attacks in Generative Diffusion Models

Backdoor attacks aim to introduce hidden behaviors into machine learning models that remain dormant under normal conditions and activate only in specific circumstances. While many early

approaches relied on explicit triggers, prior work has shown that malicious behavior can be embedded without visible signals or label inconsistencies. In particular, label-consistent backdoor attacks demonstrate that models can internalize unintended behaviors through subtle statistical manipulations of the training data, allowing poisoned samples to blend naturally into the dataset while still influencing inference-time behavior [17].

These observations extend naturally to generative models, which differ from classifiers in that they learn full data distributions rather than discrete decision boundaries. As a result, generative models may memorize or amplify rare patterns present in the training data, making them susceptible to targeted manipulation when data for specific concepts is limited. This vulnerability is especially pronounced in text-to-image models, where individual concepts or prompts may be represented by relatively few training examples.

Recent work has demonstrated effective poisoning attacks against text-to-image diffusion models. In the *Nightshade* attack, Shan et al. [16] show that injecting fewer than one hundred carefully optimized poison images associated with a target concept can significantly corrupt a Stable Diffusion SDXL model’s outputs for that prompt. Although the poisoned images remain visually similar to benign data, the resulting model generates incorrect or nonsensical images for the targeted concept, with effects that can generalize to related prompts and, in some cases, destabilize broader model behavior.

Diffusion models are particularly vulnerable to such attacks due to common fine-tuning practices. Models such as Stable Diffusion are frequently adapted using small, domain-specific datasets to learn new artistic styles or personalized concepts. Prior research indicates that fine-tuning can unintentionally alter previously learned behaviors, including safety alignment and concept suppression, even when trained on benign data [9]. When poisoned samples are present in these small retraining datasets, their influence can be amplified, enabling subtle and persistent manipulations of the model’s learned distribution.

Together, these findings demonstrate that diffusion-based generative models are susceptible to backdoor-like behaviors and targeted poisoning without requiring explicit triggers. This provides important context for steganography-based poisoning approaches, which further reduce detectability by embedding malicious signals directly into the visual content of training images.

2.3 Steganography and Watermarking in Images

Steganography and digital watermarking focus on embedding information into images while maintaining visual imperceptibility. These techniques exploit properties of human perception and signal processing to hide data in ways that are difficult for humans to notice, yet detectable and extractable by algorithms.

Image steganography aims to embed secret messages in cover images such that the existence of the message is not apparent, prioritising imperceptibility and capacity. In contrast, digital watermarking is often designed for copyright protection and authenticity verification, requiring robustness against common attacks such as compression or noise while still remaining largely invisible to a casual observer [18].

Traditional methods, such as least significant bit modification in the spatial domain and transform-domain techniques like the discrete cosine transform (DCT) or wavelet transforms, balance imperceptibility and robustness according to application needs. These approaches highlight fundamental trade-offs: spatial domain methods tend to offer high imperceptibility but low robustness, while frequency domain methods achieve greater resilience at the cost of more complex embedding and extraction [2].

In recent years, deep learning has been applied to both steganography and watermarking to improve performance, embedding capacity, and resistance to detection. Surveys of deep learning-based methods show that neural networks can learn sophisticated embedding and extraction functions, adapting to a wider range of distortions and attacks than traditional approaches [18].

Finally, hybrid methods that combine multiple transforms or neural components aim to optimise both robustness and imperceptibility, indicating an active research direction that balances these competing objectives for real-world multimedia protection and covert communication [2].

2.4 Adversarial Perturbation for Image Protection

A distinct line of research addresses the protection of images and artistic styles from unauthorised AI-based manipulation by introducing adversarial perturbations into the images themselves. Unlike traditional watermarking, these methods do not embed a recoverable signal - instead they deliberately corrupt the latent representation of the image inside a diffusion model’s VAE encoder, making the image resistant to AI editing or style imitation while remaining visually unchanged to a human observer.

Salman et al. [13] introduced PhotoGuard, which computes pixel-level perturbations by back-propagating through the frozen VAE encoder to find changes that maximally disrupt the encoded latent representation. The encoder attack variant minimises the distance between the perturbed image’s latent and a fixed target pattern far from the natural image manifold, effectively making the image uneditable by any model that uses the same VAE.

Shan et al. [15] introduced Glaze, which applies a similar adversarial perturbation strategy specifically to protect artistic style from imitation by text-to-image models. Rather than disrupting editing, Glaze shifts the perceptual representation of an artwork so that models trained on it learn a distorted version of the artist’s style. Glaze demonstrates that even small, nearly imperceptible perturbations applied before training can meaningfully alter what a model learns from an image.

Both methods share the core insight that adversarial perturbations computed against the VAE encoder survive the latent compression step and influence model behaviour - an insight that is directly relevant to the poisoning scenario explored in this thesis. The present thesis adapts this mechanism for a different purpose: rather than protecting individual images from editing or style imitation, the perturbations are applied to an entire training dataset to poison style-specific LoRA fine-tuning. The specific contributions are the evaluation of the encoder attack as a poisoning mechanism, its systematic comparison against classical watermarking methods, and an empirical analysis of how poison ratio affects deterioration effectiveness.

2.5 Steganography-Based Poisoning of Diffusion Models (Parasite)

Recent work combines steganographic embedding with data poisoning to manipulate diffusion models during fine-tuning. In particular, the *Parasite* framework demonstrates that imperceptible steganographic signals embedded in training images can alter a model’s behavior, highlighting a direct intersection between steganography, watermarking, and generative model poisoning [4].

Parasite is a backdoor attack specifically designed for diffusion models. Unlike traditional backdoor attacks that rely on visible triggers, Parasite embeds secret information in a minimally perceptible manner, using either transform-domain steganography or adversarial embedding. Only a small fraction of the fine-tuning dataset is poisoned, yet the model learns a hidden mapping that can be activated under specific conditions during inference.

The framework consists of three main steps. First, a secret message or pattern is encoded into training images using steganographic techniques that minimally alter pixel values. Second, the encoded images are inserted into a fraction of the fine-tuning dataset, often fewer than 5–10% of images. Finally, during fine-tuning, the diffusion model internalizes correlations between the hidden signals and the target output, creating a latent backdoor while behaving normally for unpoisoned inputs.

Parasite uses two primary embedding methods. Transform-domain embedding places the secret signal in the frequency domain, such as discrete cosine transform (DCT) coefficients, ensuring minimal perceptual distortion. Adversarial embedding uses gradient-based optimization to find perturbations that encode the secret while remaining imperceptible and resistant to naive detection.

The poisoned model reproduces target outputs when prompted with inputs containing hidden patterns. For example, a model fine-tuned on a poisoned dataset containing a specific art style may output style-specific features or reproduce an embedded watermark. The model maintains normal behavior for unpoisoned inputs, making the attack stealthy and difficult to detect. Empirical evaluation shows that even small poisoned subsets effectively implant the backdoor, embeddings remain functional under transformations such as JPEG compression, and the visual fidelity of outputs is preserved.

Parasite demonstrates that generative diffusion models are highly sensitive to subtle, structured perturbations in fine-tuning datasets. This provides a direct foundation for exploring steganographic watermarking for style-specific poisoning: by embedding hidden signals in a retraining dataset, one could manipulate style output, reproduce watermarks, or degrade model behavior in a controlled manner, which aligns with the goals of this thesis.

Chapter 3

Theoretical Background

This chapter provides the theoretical foundations underlying the experimental work presented in this thesis. It begins with an overview of diffusion-based generative models and explains how latent diffusion models differ from their pixel-space predecessors by introducing a compression step through a Variational Autoencoder. It then covers Low-Rank Adaptation, the parameter-efficient fine-tuning method used throughout the experiments, and concludes with an explanation of DWT-DCT watermarking, a well-established classical technique widely used in digital copyright protection, which is evaluated here as a candidate poisoning mechanism precisely because of its prevalence. The remaining watermarking methods evaluated in this thesis - a color channel shift and an adversarial VAE perturbation - are sufficiently self-contained to be introduced directly in Chapter 6 without requiring dedicated theoretical background.

3.1 Diffusion Models

Diffusion models are a class of generative models that learn to produce data by reversing a gradual noising process. The core idea originates from non-equilibrium thermodynamics and was formalized for image generation by Ho et al. [6]. In their original form, diffusion models operate directly in pixel space, meaning the forward and reverse processes both run on the full image. For a 512×512 image this involves vectors of over 786,000 values at every timestep, which is computationally expensive and limits practical applicability.

3.2 Latent Diffusion Models and the Variational Autoencoder

Latent diffusion models [12], of which Stable Diffusion is the primary example, solve this problem by moving the diffusion process out of pixel space entirely. Instead of adding and removing noise on the raw image, they first compress the image into a much smaller latent representation using a Variational Autoencoder, run the full diffusion process in that compressed space, and only decode back to pixels at the very end. This makes high-resolution image generation feasible on consumer hardware without sacrificing output quality.

3.2.1 Variational Autoencoder

A VAE [10] consists of an encoder $\mathcal{E}$ and a decoder $\mathcal{D}$. Given an input image $\mathbf{x} \in \mathbb{R}^{H \times W \times 3}$, the encoder maps it to a latent representation:

$$\mathbf{z} = \mathcal{E}(\mathbf{x}), \quad \mathbf{z} \in \mathbb{R}^{h \times w \times c} \quad (3.1)$$

where $h = H/f$, $w = W/f$, and f is the spatial downsampling factor. In Stable Diffusion v1.5, $f = 8$ and $c = 4$, so a 512×512 image is encoded to a $64 \times 64 \times 4$ latent - 48 times smaller than the original. The decoder $\mathcal{D}$ reconstructs the image from the latent: $\tilde{\mathbf{x}} = \mathcal{D}(\mathbf{z})$. The diffusion process is then applied entirely within this latent space, and the UNet denoiser never sees raw pixels at any point during training or inference.

3.2.2 Relevance to Watermark Survival

The VAE encoder is not a neutral compression step - it is a learned perceptual filter trained to discard information that is not semantically meaningful. When a steganographic watermark is embedded in pixel space, its signal occupies mid-to-high frequency components of the image. During encoding, the VAE compresses the 512×512 input to a 64×64 latent, an 64-fold reduction in spatial resolution, and in doing so attenuates precisely the frequency range where classical watermarks such as DWT-DCT embed their signal.

The practical consequence is that a watermark which is clearly measurable in pixel space may leave no trace in the latent representation that the UNet actually trains on. This creates a fundamental challenge for steganography-based poisoning of latent diffusion models: the watermark must survive not only human inspection but also the compression bottleneck of the VAE before it can influence the training process at all.

3.2.3 Why This Matters for the Thesis

This is why the thesis does not simply assume that an invisible watermark will affect the model. Instead, the experiments test several different embedding approaches and measure whether any effect actually reaches the model outputs. The DWT-DCT watermark is tested precisely because it is a classical method that operates in the frequency domain, making it a good candidate for studying whether such signals can survive the VAE compression step. The results of this investigation are presented in Chapter 6.

3.3 Low-Rank Adaptation

Full fine-tuning of large generative models requires updating billions of parameters, which is computationally expensive on consumer hardware and risks overwriting previously learned behaviors. Low-Rank Adaptation (LoRA), introduced by Hu [7], offers a practical alternative: the pretrained model weights are frozen entirely, and a small number of additional trainable parameters are injected into selected layers. Only these injected parameters are updated during training.

3.3.1 Formulation

The key insight behind LoRA is that when a model is fine-tuned on a specific style, the meaningful changes to its weights tend to be concentrated in a small number of directions rather than spread across the full parameter space. Rather than learning a full $d \times k$ update matrix, LoRA decomposes it into two smaller matrices:

$$\mathbf{W} = \mathbf{W}_0 + \mathbf{B}\mathbf{A} \tag{3.2}$$

where $\mathbf{A} \in \mathbb{R}^{r \times k}$ and $\mathbf{B} \in \mathbb{R}^{d \times r}$, with rank $r \ll \min(d, k)$. The original weights $\mathbf{W}_0$ are frozen throughout training. The rank r controls how much the adapter can express - a higher rank allows more complex style changes but requires more memory and risks overfitting on small datasets.

3.3.2 Why LoRA Was Chosen for This Thesis

LoRA is the natural choice for style-specific poisoning experiments for four reasons. First, it is computationally feasible on a single consumer GPU, allowing many experiments to be run with different hyperparameters and watermarking configurations. Second, it preserves the base model entirely, meaning the same Stable Diffusion v1.5 checkpoint is reused across all experiments and any behavioral differences between runs are attributable solely to the training data. Third, the adapter parameters are small and self-contained, which makes them a well-defined target for analyzing what the model learned from a poisoned dataset. Fourth, because all learned changes are concentrated in a small number of parameters, poisoned training data has a more direct influence on the result - making it easier to both introduce and measure the poisoning effect.

3.3.3 Application to UNet Attention Layers

In Stable Diffusion, LoRA adapters are injected into the attention projection matrices of the UNet denoiser - the query, key, value, and output projections - since these layers mediate how the model relates visual features to each other and to the conditioning text. Optionally, feed-forward and convolutional layers can also be targeted to give the adapter greater capacity to reshape the model's style representation. The choice of which layers to include is itself a hyperparameter explored in the model selection phase described in Chapter 5.

3.4 DWT-DCT Image Watermarking

The watermarking method employed in this thesis combines the Discrete Wavelet Transform (DWT) and the Discrete Cosine Transform (DCT) to embed a signal into the frequency domain of an image. This hybrid approach, proposed by Al-Haj [1], achieves a good balance between imperceptibility and robustness by embedding the signal in a frequency range that is difficult for the human eye to detect.

Since the VAE is known to suppress mid-frequency signals, a classical frequency-domain watermark like DWT-DCT is not expected to survive the encoding step. This thesis tests this directly rather than assuming it, using DWT-DCT as a baseline that represents the weaker end of the poisoning spectrum. The results are presented in Chapter 6.

3.4.1 Discrete Wavelet Transform

The DWT decomposes an image into multi-resolution subbands by applying pairs of low-pass and high-pass filters along rows and columns. A single-level DWT of an image I produces four subbands:

- cA — approximation subband (low-frequency content, coarse structure)
- cH , cV , cD — detail subbands (horizontal, vertical, and diagonal high-frequency edges)

The approximation subband cA captures the perceptually dominant content of the image, while the detail subbands encode fine structure. Embedding in cA therefore affects the content that the human visual system is most sensitive to, which is why the embedding strength must be carefully controlled.

3.4.2 Discrete Cosine Transform Embedding

After computing the DWT, the approximation subband cA is divided into non-overlapping 8×8 blocks, and the DCT is applied to each block. The DCT transforms the block into frequency coefficients arranged from low to high frequency. A single bit $b \in \{0, 1\}$ of the watermark is embedded by modifying a mid-frequency coefficient C_{ij} of each block:

$$C'_{ij} = \begin{cases} C_{ij} + \alpha & \text{if } b = 1 \\ C_{ij} - \alpha & \text{if } b = 0 \end{cases} \quad (3.3)$$

where α is the embedding strength parameter. The inverse DCT and inverse DWT are then applied to reconstruct the watermarked image. The process is repeated independently for each color channel and the extracted bits are averaged across channels to improve robustness.

Chapter 4

Experimental Setup and Methodology

This chapter describes the experimental infrastructure constructed for this thesis. It covers the dataset used for fine-tuning, the selection and justification of the base generative model and the fine-tuning approach.

4.1 Dataset

The ultimate goal of the fine-tuning process is to produce a model that generates images in the style of Van Gogh regardless of the input prompt - a recognisable and well-defined artistic style that makes evaluation straightforward and visually unambiguous. Van Gogh was chosen deliberately for this purpose. His style is distinctive enough that even a casual observer can identify it, which makes human and automated evaluation of style transfer more reliable. Additionally, given that this thesis investigates mechanisms that could be used to protect artists from unauthorised model training, it would be contradictory to conduct the experiments using the work of a living artist without consent. Van Gogh’s works are in the public domain, making them an ethically appropriate choice for research of this nature.

The dataset used for all fine-tuning experiments in this thesis is a collection of Van Gogh paintings sourced from Kaggle [8]. The original dataset contains 2025 images, however a significant portion consisted of low-resolution scans, and works that either did not belong to Van Gogh’s recognisable style or bore little resemblance to it, all of which would introduce noise into the training process and dilute the stylistic signal the model is expected to learn. After filtering out these unsuitable images, 559 images were retained, spanning a wide range of Van Gogh’s works including landscapes, portraits, and still life paintings, providing sufficient stylistic diversity to train a meaningful style adapter.

Raw images in the retained set vary in resolution, aspect ratio, and color mode. To ensure compatibility with Stable Diffusion v1.5, which expects square RGB inputs of 512×512 pixels, all images were preprocessed through a standardised pipeline. Each image was first converted to RGB to handle any grayscale or palette-mode files, then center-cropped to a square aspect ratio to preserve the central content of each painting, and finally resized to 512×512 pixels. The processed images were saved to a dedicated directory and used as the fixed input for all subsequent training runs.

Center-cropping was chosen over padding or stretching because it preserves the natural proportions and brushstroke characteristics of each painting, which are the primary stylistic features the model is expected to learn. Stretching would distort these features, and padding would introduce large uniform regions that carry no stylistic information and could mislead the training process.

4.2 Base Model Selection

Stable Diffusion v1.5 [12], released by RunwayML, was selected as the base model for all experiments in this thesis. It is a latent diffusion model trained on a large corpus of image-text pairs and is capable of generating high-quality images from text prompts. Several factors motivated this choice.

First, Stable Diffusion v1.5 is lightweight enough to be fine-tuned on a single consumer GPU, which was a practical constraint of this work. Larger and more capable models such as Stable Diffusion XL or newer architectures would require significantly more memory and compute, making the scale of hyperparameter search conducted in Chapter 5 infeasible.

Second, it is one of the most widely used and documented text-to-image models in the research community, with extensive tooling, well-understood behavior, and a large number of existing studies on fine-tuning and poisoning. This makes results obtained on this model interpretable and comparable to related work.

Third, the model’s architecture - a UNet denoiser operating in the latent space of a pretrained VAE - is well suited to LoRA-based adaptation, as the attention layers of the UNet are natural injection points for low-rank adapters.

The model was downloaded from the Hugging Face Hub under the identifier *runwayml/stable-diffusion-v1-5* and stored locally to ensure that all experiments used an identical, fixed checkpoint. No modifications were made to the base model weights at any point.

To verify the base model before any fine-tuning, it was prompted with the single word *cat*. This prompt was chosen as a consistent test case throughout the thesis because a cat is a recognisable subject with well-defined visual characteristics, yet varied enough in pose, color, and composition to produce meaningfully different results across different random seeds. As expected, the unmodified base model produced images of normal cats with no stylistic influence from Van Gogh. This confirmed that any stylistic shift observed in later experiments is attributable to the fine-tuning process rather than the base model itself, and established the need for the adaptation described in Chapter 5.



FIGURE 4.1: Example images generated by the unmodified Stable Diffusion v1.5 base model using the prompt *cat*. The outputs show no stylistic influence from Van Gogh, confirming the base model as a neutral starting point for all fine-tuning experiments.

4.3 Fine-Tuning Approach

As established in Chapter 3, full fine-tuning of Stable Diffusion v1.5 would require updating the entirety of the model’s parameters, which is both computationally expensive and risks destroying previously learned knowledge through catastrophic forgetting. Low-Rank Adaptation was therefore chosen as the fine-tuning method for all experiments in this thesis.

The LoRA adapter was applied exclusively to the UNet denoiser, leaving the VAE and the text encoder entirely frozen throughout training. The VAE is frozen because its role is image compression and reconstruction rather than style generation, and modifying it would risk destabilising the latent space that all experiments depend on. The text encoder is frozen because the goal is unconditional style transfer - the model should produce Van Gogh-style outputs regardless of the input prompt, not only when prompted with style-specific language.

Only the UNet receives trainable adapter weights, injected into its attention and optionally feed-forward and convolutional layers. The specific set of layers targeted is a hyperparameter that was varied during the model selection phase described in Chapter 5.

Training was performed using the standard diffusion objective: the adapter weights are updated to minimise the mean squared error between the noise predicted by the UNet and the actual noise added during the forward diffusion process. All training images were encoded into the latent space using the frozen VAE encoder before being passed to the UNet, meaning the adapter learns entirely from latent representations rather than raw pixels. This is the step at which watermark survival becomes relevant, as discussed in Chapter 3.

Chapter 5

Baseline Model Training and Selection

This chapter describes the process of training and selecting a Van Gogh style LoRA adapter to serve as the clean baseline for all poisoning experiments. It covers the initial failed attempts, the discovery of the correct training strategy, the hyperparameter search, and the final model selection.

5.1 Initial Experiments

With the fine-tuning approach established, the first training attempts were conducted to assess how well the model could learn the Van Gogh style. An initial adapter was trained using rank $r = 16$ over 20 epochs, targeting only the attention projection layers of the UNet. The resulting model showed no meaningful stylistic shift - generated images of cats were visually indistinguishable from those produced by the unmodified base model, with no trace of Van Gogh’s characteristic brushwork or color palette.

The rank was subsequently increased to $r = 64$, which produced a marginal improvement. Some images began to show faint painterly qualities and a warmer color palette reminiscent of Van Gogh’s work, but the effect was inconsistent across different seeds and too subtle to be considered a successful style transfer. It was clear that rank alone was not the limiting factor, and that a more fundamental change to the training setup was needed.

5.2 Training Prompt Discovery

Early training runs conditioned the model on the text prompt *painting in the style of Van Gogh*, following the conventional approach of associating a style with a descriptive phrase. However this produced disappointing results - the model learned to associate the Van Gogh style only with that specific phrase rather than internalising it unconditionally. When prompted with *cat* at inference time, the style had little influence on the output because the adapter had learned to activate only in response to the training prompt.

The key insight was to remove the prompt entirely during training, conditioning the model on an empty string instead. The reasoning is straightforward: since the goal is for every generated image to carry Van Gogh’s style regardless of what the user prompts, the model should not be taught to associate the style with any particular text. By training on an empty prompt, the stylistic signal from the training images is absorbed into the adapter weights unconditionally, and the style becomes the default output behaviour rather than a text-triggered response. This is demonstrated throughout the thesis by prompting the model with the single word *cat* at inference time - a subject the model was never shown during training - to verify that the style transfer is truly unconditional.

This change produced an immediate and dramatic improvement. With only 3 epochs of training at $r = 64$, the generated images clearly exhibited Van Gogh’s characteristic swirling brushstrokes, bold colors, and impasto texture. Encouraged by this result, training was extended to 20 epochs to see whether further training would strengthen the effect. Contrary to expectation, the 20-epoch model performed worse - a large proportion of generated images became desaturated or fully black and white, and in some cases the subject matter became unrecognisable. This indicated that 3 epochs was close to the optimal training duration for this dataset size, and that over-training caused the adapter to overfit and degrade generation quality rather than improve it.

5.3 Hyperparameter Search

Given the sensitivity of results to training configuration observed in the initial experiments, a systematic hyperparameter search was conducted to identify the optimal combination of parameters. Rather than tuning one parameter at a time, a full grid search was performed over the following ranges:

- **Epochs:** 1, 5, 10 — the number of full passes through the training dataset. More epochs mean more exposure to the training style but risk overfitting, as observed in the initial experiments where 20 epochs degraded quality compared to 3. A range of low values was chosen deliberately to explore the region where style transfer is effective without over-training.
- **Rank (r):** 8, 16, 32, 64 — controls the capacity of the LoRA adapter. Higher rank allows the adapter to express more complex style changes but increases memory usage and the risk of overfitting on a small dataset. The range was chosen to span from minimal capacity at $r = 8$ to the highest value feasible on a consumer GPU at $r = 64$, covering the full practical spectrum.
- **LoRA alpha (α):** 32, 128 — a scaling factor that controls how strongly the adapter influences the model’s output. The effective scaling is determined by the ratio α/r , where a higher ratio means the adapter’s learned changes have more weight relative to the frozen base model. Two values were selected to compare a weaker influence ($\alpha = 32$) against a stronger one ($\alpha = 128$), to see whether the model benefits from a subtle nudge or a more forceful style push.
- **Learning rate:** 10^{-4} , 5×10^{-5} , 10^{-5} — controls how much the adapter weights change at each training step. 10^{-4} is a standard starting point for LoRA fine-tuning recommended in the original LoRA paper [7] and widely used in practice. 5×10^{-5} is a commonly recommended value specifically for style transfer tasks where more cautious updates help preserve image quality. 10^{-5} was included as a conservative lower bound to test whether very slow learning produces better or worse results with a small dataset.
- **Target modules:** attention only ($\mathbf{W}_q, \mathbf{W}_k, \mathbf{W}_v, \mathbf{W}_o$); attention and feed-forward layers - determines which parts of the UNet receive trainable adapter weights. Attention-only is the minimal and most widely used configuration. Including feed-forward layers gives the adapter greater capacity to reshape the model’s internal representations and was included to test whether the additional parameters meaningfully improve style transfer at the cost of longer training.

Every combination of these parameters was trained, resulting in 144 distinct models in total. Each model was then used to generate 20 images using the prompt *cat*, producing a standardised

output set that could be evaluated consistently across all configurations. The generated images were scored using the CLIP-based evaluation methodology described in the following section, and the scores were used to rank all models and identify which parameter combinations consistently produced the best results.

5.4 Evaluation Methodology

To select the best model objectively rather than by visual inspection alone, a composite CLIP-based score was computed for each model. CLIP, introduced by Radford et al. [11] and pretrained on 400 million image-text pairs, maps both images and text into a shared embedding space where cosine similarity reflects semantic relatedness. Hessel et al. [5] demonstrated that this similarity correlates strongly with human judgement, making it suitable for reference-free evaluation without requiring ground truth images.

Two prompts were used to score each generated image:

- **Content score** — CLIP similarity between the image and “*a cat*”, measuring how well the model preserved the subject matter of the prompt.
- **Style score** — CLIP similarity between the image and “*a painting by Van Gogh with swirling brushstrokes and bold colors*”, measuring how strongly the Van Gogh aesthetic is present in the output.

The combined score was computed as the simple average of the two. This composite metric ensures that a model is only rewarded if its output is simultaneously cat-like and Van Gogh-like. Model that generates stylistically correct images with an unrecognisable subject, or perfectly recognisable cats with no stylistic influence, would score lower than one that achieves both. This directly reflects the goal of the fine-tuning process and provides a single comparable number for ranking all models in the hyperparameter search.

5.5 Results and Model Selection

Models trained for 1 and 10 epochs both performed poorly across all parameter combinations. The 10-epoch models scored consistently in the range of 22.4 to 22.7, while 1-epoch models scored even lower on average. Crucially, the highest-scoring models from both of these epoch counts scored below the lowest-scoring models trained for 5 epochs, indicating that 5 epochs represents a meaningful threshold for effective style transfer with this dataset size and configuration.

This result aligns with the known behaviour of LoRA fine-tuning on small datasets. When the training set is limited in size, the adapter rapidly absorbs the dominant stylistic patterns within the first few epochs, after which continued training causes it to overfit to the specific images rather than generalising the style. Hu et al. [7] demonstrate that the low-rank constraint inherently limits the number of independent directions the adapter can learn, which means the useful signal is captured early and further training primarily reinforces dataset-specific artefacts rather than broadening the learned representation. This effect is amplified in style transfer scenarios where all training images share a single artist’s aesthetic - the stylistic signal is strong and consistent, so the adapter converges quickly, and additional epochs push it past the point of generalisation into memorisation.

Among the 5-epoch models, combined scores ranged frequently between 23 and 26, with occasional results reaching 26 or above. The top five models by combined score are shown in Figure 5.1.

The best individual model achieved a content score of 28.521, a style score of 26.942, and a combined score of 27.731, using the configuration $r = 64$, $\alpha = 32$, learning rate 5×10^{-5} , targeting attention and feed-forward layers.

Examining the top ten models revealed several consistent patterns. The rank $r = 64$ appeared in every top-performing model, confirming that higher adapter capacity is beneficial for this task. The value $\alpha = 32$ dominated over $\alpha = 128$ across nearly all configurations, which is consistent with the earlier observation that over-training degrades quality - a lower alpha applies the adapter’s changes more conservatively, which appears to produce better generalisation on a small dataset. The learning rate of 5×10^{-5} was the most common among top performers, though with less variance than the other parameters, suggesting it is a safe default rather than a critical differentiator.

A consistent pattern observed across the entire search was that content scores were higher than style scores in virtually every model evaluated, which is clearly visible in Figure 5.1. This suggests that preserving subject recognisability is easier for the adapter than fully capturing Van Gogh’s distinctive visual style. The gap narrowed in the better-performing models, indicating that higher combined scores reflect genuine improvement in style transfer rather than simply better subject fidelity.

Given that 5 epochs produced the best results but training time is a practical concern for the large number of poisoning experiments to follow, an additional set of models was trained using 3 epochs with the most consistently successful parameter combinations, and with the addition of convolutional layers included in the target modules. The model *van-gogh-lora-e3-r64-a32-d0.05-lr5e-05-ga4-attn+ff+conv* achieved a combined CLIP score of 27.047, slightly below the best 5-epoch model but trained in significantly less time. This model was selected as the clean baseline for all subsequent poisoning experiments, as the marginal score difference does not justify the additional training cost and the shorter training time makes it far more practical for repeated use across many watermarking configurations.

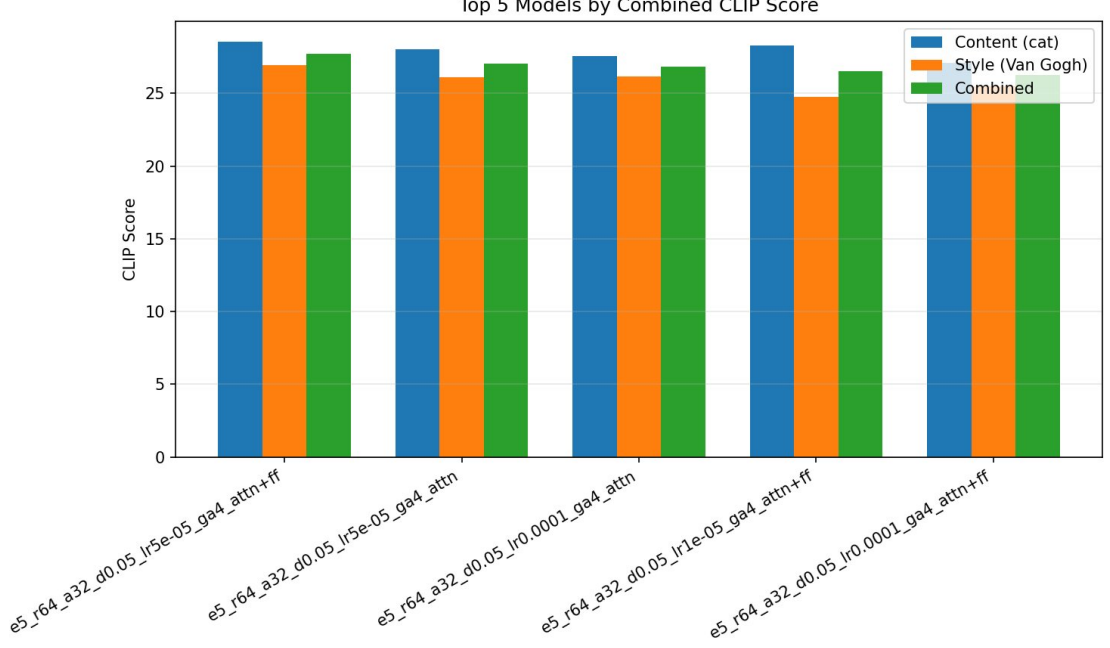


FIGURE 5.1: Top 5 models by combined CLIP score from the hyperparameter search. All five use $r = 64$ and $\alpha = 32$, confirming these as the dominant parameters. Content scores consistently exceed style scores across all models. *attn* denotes adapters targeting only the attention projection layers ($\mathbf{W}_q, \mathbf{W}_k, \mathbf{W}_v, \mathbf{W}_o$), while *attn+ff* additionally includes the feed-forward layers of the UNet.

5.6 Qualitative Comparison with Public Models

To contextualise the quality of the selected baseline model, its outputs were compared against those produced by a publicly available Van Gogh style image generator. The DeepAI text-to-image API¹ was selected for this comparison as it is a widely accessible and well-known public tool that offers Van Gogh style generation without requiring specialised setup, making it a representative benchmark for what a non-specialist user might use to imitate an artist’s style. The baseline model was prompted with *cat* - its standard generation prompt throughout this thesis - while the public model required the explicit style instruction *cat in the style of Van Gogh* to produce any stylistic output at all. This difference in prompting is itself a meaningful observation: the baseline model applies the Van Gogh style unconditionally to any prompt, while the public model requires the user to explicitly request it. The results are shown in Figure 5.2.

The comparison reveals a notable difference in approach. The publicly available model appears to apply a superficial stylistic overlay - producing cats placed against a blue swirling background reminiscent of *Starry Night* - rather than genuinely internalising Van Gogh’s visual language. The brushwork, colour palette, and compositional style of the subject itself remain largely unaffected. The baseline model produced in this thesis, by contrast, applies the style unconditionally and pervasively - the cats themselves are rendered in Van Gogh’s characteristic impasto texture and bold colours, with no single iconic work dominating the output. This distinction validates the training approach: by using an empty prompt and training on a diverse dataset of Van Gogh’s works rather than relying on prompt engineering, the adapter learns a deeper and more general representation of the style rather than a surface-level association with a single famous painting.

¹<https://deepai.org/machine-learning-model/text2img>



FIGURE 5.2: Comparison of cat images generated by the baseline Van Gogh LoRA adapter trained in this thesis using the prompt *cat* (top) and the DeepAI public Van Gogh style generator using the prompt *cat in the style of Van Gogh* (bottom). The public model applies a superficial overlay based on *Starry Night*, while the baseline model pervasively applies Van Gogh’s style to the subject itself without requiring an explicit style prompt.

5.7 Final Experiment Setup

With the baseline model established, a fixed and fully reproducible experiment pipeline was designed to ensure all poisoning experiments are conducted under identical conditions. The pipeline is implemented as a single script that accepts a watermarking algorithm and a set of parameters, and executes three sequential steps automatically.

Step 1 — Dataset Watermarking. The preprocessed dataset of 559 images is passed through the selected watermarking algorithm. A configurable poison ratio determines what fraction of images receive the watermark, while the remaining images are copied unchanged to keep the dataset size constant. Unless specified otherwise, a poison ratio of 100% is used, meaning all training images are watermarked. The watermarked dataset is saved to a dedicated directory named after the algorithm and its parameters, so the same watermarked data can be reused across multiple runs without re-embedding.

Step 2 — LoRA Training. A fresh LoRA adapter is trained on the watermarked dataset using the fixed hyperparameters established during model selection: rank $r = 64$, $\alpha = 32$, learning rate 5×10^{-5} , 3 epochs, gradient accumulation over 4 steps, targeting attention, feed-forward, and convolutional layers. The base Stable Diffusion v1.5 checkpoint is loaded from the local copy, the VAE and text encoder are frozen, and only the injected adapter weights are updated. The trained adapter is saved to a dedicated experiment directory upon completion.

Step 3 — Image Generation. The trained adapter is loaded on top of the base model and used to generate images using the prompt *cat*. Two sets of images are produced: 100 images generated with seeds offset from the global seed, and 10 images generated with fixed seeds from 1 to 10. The fixed-seed images are identical in starting noise across all experiments, meaning any pixel-level or perceptual differences between them reflect only the influence of the watermarked training data on the adapter. These images are compared against the corresponding fixed-seed

outputs of the empty baseline experiment, which serves as the reference point for all Group 2 deterioration metrics in Chapter 6.

To ensure full reproducibility across all runs, all sources of randomness in the training pipeline are controlled using a global seed of 42. This includes the DataLoader shuffle order, the noise sampled at each training step, and the diffusion timestep selection. The dataset file list is also sorted before loading to remove any dependency on filesystem ordering. Together these measures ensure that two runs of the same experiment with the same watermarked dataset produce highly reproducible adapter weights with negligible run-to-run variation under identical hardware conditions, making the results directly comparable across different watermarking configurations. Full byte-level determinism cannot be guaranteed however, as parallel GPU floating point operations may execute in different orders depending on hardware load, introducing minor numerical differences even when all random seeds are fixed.

Chapter 6

Watermarking and Poisoning Experiments

This chapter presents the main experimental contribution of the thesis. Having established a clean baseline model and a reproducible experiment pipeline in the previous chapter, four watermarking strategies of varying aggressiveness are applied to the Van Gogh training dataset. For each strategy, a LoRA adapter is trained on the poisoned dataset under identical conditions and the resulting model is evaluated using a two-group scoring framework. The chapter begins by defining the evaluation metrics used throughout, then presents each watermarking method and its results individually, and concludes with a combined comparison across all methods to identify which approach best achieves the thesis goal of invisible yet effective poisoning.

6.1 Evaluation Framework

It is worth acknowledging upfront that the metrics used in this chapter, while academically grounded, are ultimately supporting tools rather than the primary evidence. Both the presence of a watermark in the training images and the deterioration of the model’s output are phenomena that are immediately apparent to the naked eye - a large red square needs no PSNR score to be identified as visible, and a generated image covered in red squares needs no CLIP drift calculation to be identified as deteriorated. The metrics serve to quantify these observations precisely, enable comparison across experiments, and provide reproducible numerical evidence for the thesis. However the visual output of each experiment remains the most direct and interpretable indicator of whether a watermarking strategy succeeded or failed, and the scores should be read in that context.

Before presenting individual results, this section defines the two groups of metrics used to evaluate every experiment. The first group measures how much the watermark changed the training images relative to the originals. The second group measures how much the resulting model drifted from the empty baseline. Together they capture the central trade-off of this thesis: a useful watermark should be invisible in the first group and maximally effective in the second.

6.1.1 Group 1 — Watermark Visibility

Group 1 metrics compare all 559 watermarked training images against their original counterparts pixel by pixel. A high score indicates the watermark is imperceptible; a low score indicates it is visibly destructive.

PSNR (Peak Signal-to-Noise Ratio) measures the difference between the watermarked and original image in decibels. In the watermarking literature, values above 40 dB are widely considered imperceptible to the human eye, while values below 30 dB indicate clearly visible distortion [18, 1]. Higher is better for invisibility. It is worth noting that even the empty baseline produces a PSNR of approximately 32 dB rather than infinity - this is because opening, processing, and saving an image introduces minor rounding differences at the pixel level even when no intentional modification is made, a known artefact of image I/O pipelines.

SSIM (Structural Similarity Index) measures whether the watermarked image preserves the structural patterns, textures, and luminance of the original [18]. It ranges from 0 to 1, where 1 means pixel-identical. Higher is better for invisibility. As with PSNR, a perfect score of 1.0 is not expected even for the empty baseline, since the same image I/O rounding differences that affect PSNR also cause minor structural deviations that prevent a perfect SSIM score. The empty baseline therefore serves as the practical ceiling for Group 1 scores rather than the theoretical maximum.

6.1.2 Group 2 — Model Deterioration

Group 2 metrics evaluate the generated images from the trained model and compare them against images generated by the empty baseline model (Section 6.2) using identical seeds. A high deterioration score indicates the watermark successfully shifted the model’s behaviour. It is important to note that the metrics in this group are split into two subsets. The CLIP Combined, Brightness, and Sharpness scores are computed over all 110 generated images - 100 with offset seeds and 10 with fixed seeds - providing a broad picture of the model’s overall output distribution. The SSIM vs Empty and CLIP Drift vs Empty scores are computed exclusively over the 10 fixed-seed images, since only these have a direct counterpart in the empty baseline to compare against.

The empty baseline experiment - where no watermark is applied - serves as the reference for all Group 2 comparisons. This ensures that all experiments are measured relative to a model trained through exactly the same pipeline under identical conditions, with the only variable being the watermarking applied to the training data. The empty baseline therefore produces SSIM vs Empty of 1.0 and CLIP Drift of 0.0 by definition, and any deviation from these values in other experiments reflects the influence of the watermark on the trained model.

CLIP Combined is the average of a content score and a style score, both computed using CLIP similarity. The content score measures how much the generated image resembles *a cat*, and the style score measures how much it resembles *a painting by Van Gogh with swirling brushstrokes and bold colors*. This metric is used as a sanity check - a large drop relative to the empty baseline would indicate the model’s outputs have degraded significantly in quality or style fidelity.

Brightness and **Sharpness** are auxiliary sanity metrics. Brightness is the average pixel intensity across the generated image, useful for detecting whether the model has begun generating unusually dark or washed-out outputs. Sharpness is the variance of the Laplacian filter applied to the image, measuring how much fine edge detail is present.

SSIM vs Empty is the primary deterioration metric. It compares the seeded outputs of the experiment model against the seeded outputs of the empty baseline at identical seeds. A lower value means the poisoned model produces images that are structurally more different from the empty baseline.

CLIP Drift vs Empty is the second primary deterioration metric. It measures the cosine distance between CLIP image embeddings of the experiment’s seeded outputs and the empty

baseline’s seeded outputs. A higher value means the model’s outputs have drifted further from the empty baseline in perceptual space.

6.1.3 Ideal Score Profile

The ideal result for this thesis would be a watermark that scores similarly to the empty baseline in Group 1 - meaning it is effectively invisible to the human eye and to pixel-level metrics - while scoring similarly to the most aggressive method in Group 2, meaning the model trained on the watermarked data produces outputs that are perceptually far from the empty baseline. In short: invisible on the training images, yet impactful on the trained model.

6.2 Empty Baseline

6.2.1 Method

The empty watermark applies no modification to any training image. Each image is opened, converted to RGB, and saved to the output directory without any intentional modification to pixel values. This experiment serves two purposes. First, it acts as a sanity check for Group 2 - it establishes the deterioration scores that result from training on completely unmodified data, meaning any real watermarking method that produces similar Group 2 scores has had no measurable effect on the model’s behaviour. Second, and equally importantly, it defines the target for Group 1 - a watermarking method that achieves Group 1 scores close to the empty baseline is effectively invisible, since the modifications it introduces are indistinguishable from the minor rounding artefacts of the image I/O pipeline itself. The ideal watermarking method would match the empty baseline in Group 1 while diverging from it as much as possible in Group 2. It is worth noting that even the empty baseline produces a PSNR of approximately 32 dB rather than infinity - this is because the pipeline opens each image, processes it through a standardised RGB conversion step, and re-saves it as PNG. This encode-decode cycle introduces minor rounding differences at the pixel level even when no watermark is applied, which is why 32 dB rather than infinity is the observed baseline score.

6.2.2 Results

- **Group 1** — **PSNR:** 32.04 dB **SSIM:** 0.926
- **Group 2** — **SSIM vs Empty:** 1.0 **CLIP drift:** 0.0

As expected, the empty baseline produces Group 1 scores that reflect minor differences caused solely by the file save-and-reload cycle rather than any actual modification. These scores now serve as the reference ceiling for Group 1 and the reference point for Group 2 in all subsequent experiments - all other methods are measured relative to these values, with lower SSIM vs Empty and higher CLIP drift indicating stronger poisoning effect. Example outputs from the empty baseline model are shown in Figure 6.1, and represent what a clean, unpoisoned Van Gogh style adapter produces - serving as the visual reference against which all poisoned models are compared. The Group 2 scores of SSIM vs Empty = 1.0 and CLIP drift = 0.0 are expected by definition, since the empty baseline is compared against itself - any deviation from these values in subsequent experiments is attributable solely to the effect of the watermark on training.



FIGURE 6.1: Example images generated by the model trained on the empty baseline dataset. These represent clean, unpoisoned outputs and serve as the visual reference for all subsequent experiments.

6.3 DWT-DCT Watermark

6.3.1 Method

The DWT-DCT watermark embeds a fixed 64-bit signal into the frequency domain of each training image using the hybrid approach described in Chapter 3. The embedding strength is controlled by the parameter α , which determines how much each DCT coefficient is modified. All experiments in this chapter use $\alpha = 5.0$, which produces a watermark that is measurable in pixel space but invisible to the human eye. Unlike typical watermarking applications, no attempt is made to recover or verify the embedded bits from generated images - the sole purpose of the embedding is to introduce a structured signal into the training data and observe whether it influences the model's learned distribution.

6.3.2 Results

- **Group 1** — **PSNR:** 32.008 dB **SSIM:** 0.9257
- **Group 2** — **SSIM vs Empty:** 0.7009 **CLIP drift:** 0.0873

The Group 1 scores confirm that the DWT-DCT watermark at $\alpha = 5.0$ produces changes that are effectively invisible, with a PSNR of 32 dB - virtually identical to the empty baseline - indicating that the watermark introduces no measurable modification beyond what the encode-decode pipeline already introduces. This should not be interpreted as the watermark being imperceptible in absolute terms: a PSNR of 32 dB is below the 40 dB imperceptibility threshold. Rather, it means the watermark adds nothing on top of the processing artefacts that are unavoidable in this pipeline. The Group 2 scores tell the same story: they are close to the empty baseline reference, with SSIM vs Empty of 0.7009 compared to 1.0 for empty baseline, and CLIP drift of 0.0873 compared to 0.0. The residual difference from the empty baseline reflects minor non-determinism in GPU floating point operations rather than any watermarking effect. The watermark is invisible in Group 1 for the right reason - it is genuinely imperceptible - but it is also invisible in Group 2, which is precisely the wrong outcome. This confirms the hypothesis from Chapter 3: the DWT-DCT signal is attenuated by the VAE encoder before the UNet ever sees it, and therefore has no measurable effect on the trained model's output distribution. Example outputs from this experiment are shown in Figure 6.2 and are visually indistinguishable from the empty baseline in Figure 6.1 - the outputs maintain the same colours, style, and quality, with only minor differences in pose and composition consistent with the natural generation diversity expected from any two independently trained models. This further confirms that no meaningful poisoning occurred and that the residual Group 2 scores reflect this natural variation rather than any watermarking effect.



FIGURE 6.2: Example images generated by the model trained on the DWT-DCT watermarked dataset. The outputs are visually indistinguishable from the empty baseline, consistent with the Group 2 scores showing no measurable deterioration.

6.4 Red Square

6.4.1 Method

The red square watermark overlays a solid red rectangle of fixed size in the bottom-right corner of every training image. The square covers approximately 10% of the image area and is fully opaque, making it clearly visible to any observer. This method represents the opposite extreme from DWT-DCT - it is maximally aggressive and makes no attempt at imperceptibility. Its purpose is to establish an upper bound on poisoning effectiveness: if even a clearly visible modification of this scale produces only modest deterioration scores, it would suggest the pipeline is resistant to poisoning. Conversely, if it produces strong deterioration, it confirms the pipeline is capable of propagating training-time signals into generated outputs.

6.4.2 Results

- **Group 1** — **PSNR:** 25.591 dB **SSIM:** 0.9153
- **Group 2** — **SSIM vs Empty:** 0.5765 **CLIP drift:** 0.1866

The Group 1 scores reflect the expected visibility of the modification - a PSNR of 25.6 dB is clearly below the imperceptible threshold, and visual inspection of the watermarked training images confirms the square is immediately obvious. The Group 2 scores show clear deterioration relative to the empty baseline, with SSIM vs Empty dropping to 0.5765 compared to 1.0 for the empty baseline, and CLIP drift rising to 0.1866 compared to 0.0. Inspection of the generated images, shown in Figure 6.3, confirmed that the red square appears in virtually all outputs, sometimes varying slightly in size or color, but consistently present across different seeds and prompts.

An interesting observation is that the square occasionally appears in shades of blue or purple rather than the original red. This is likely the result of several interacting factors. First, the VAE does not encode color perfectly - the latent representation of a saturated red region may be decoded with a color shift depending on how the VAE’s internal color space is organised [12]. Second, the UNet is trained to predict noise rather than pixel colors directly [6], meaning it learns a latent-space representation of the square’s position and shape rather than its exact RGB values, and the decoded color can differ from the original. Third, the Van Gogh style adapter introduces a strong blue and indigo color bias from the training dataset, and this style signal may bleed into the square’s color during generation, shifting it toward cooler tones [7]. Rather than being a failure of the poisoning, this behaviour actually demonstrates that the model is not simply memorising and stamping exact pixel values from the training images - it is learning a latent representation of the square that interacts with both the diffusion process and the style signal in non-trivial ways.

The deterioration scores, while the highest of all three methods, are nonetheless not as extreme as one might expect from a modification this aggressive. This is explained by the nature of the output images themselves - the generated cats are still recognisably cats occupying the majority of the image, and the square covers only a small region of the output. CLIP and SSIM are computed over the entire image, so the large unchanged portion of each output moderates the overall score. The watermark has clearly propagated, but the model has not lost the ability to generate coherent subject matter - it has simply learned to consistently append the red square as an additional feature.

This confirms that the pipeline is capable of propagating a training-time signal into the model's generated outputs, but only when the signal is coarse enough to survive the VAE encoding step. From this point forward, the target for any new watermarking method is to achieve Group 1 scores matching the empty baseline - indicating true invisibility, while achieving Group 2 scores matching or exceeding those of the red square, indicating a poisoning effect as strong as the most aggressive method tested so far.



FIGURE 6.3: Example images generated by the model trained on the red square poisoned dataset. The square is consistently present in the bottom-right corner of every output.

6.5 Color Channel Shift

6.5.1 Method

The color channel shift watermark applies a uniform additive offset to a single color channel across every pixel of every training image. Unlike DWT-DCT which operates in the frequency domain, this method operates at the coarsest possible level - it shifts the global color distribution of the entire image. The motivation for this approach comes directly from the failure of DWT-DCT: since the VAE encoder preserves global color information as semantically meaningful, a color shift is expected to survive the VAE bottleneck and propagate into the latent representation that the UNet trains on.

Three channel variants were tested - red, green, and blue - each with a fixed shift of +20 pixel values, applied by adding the offset to the target channel and clamping to the valid range $[0, 255]$. All other channels are left unchanged.

6.5.2 Results — Red Channel

- **Group 1** — PSNR: 25.802 dB SSIM: 0.9145
- **Group 2** — SSIM vs Empty: 0.6424 CLIP drift: 0.0927

Shifting the red channel upward produces a warm reddish tint across the entire image. The Group 1 PSNR of 25.8 dB confirms the modification is measurable and below the imperceptible

threshold, though the visual change is subtle on Van Gogh’s naturally warm palette. The Group 2 scores show some deterioration relative to the empty baseline, but the effect is modest. A plausible explanation is that Van Gogh’s paintings are already heavily biased toward warm reds, oranges, and yellows - shifting the red channel further in that direction pushes the images deeper into a color region the VAE already knows well from the training distribution, limiting the latent disruption.

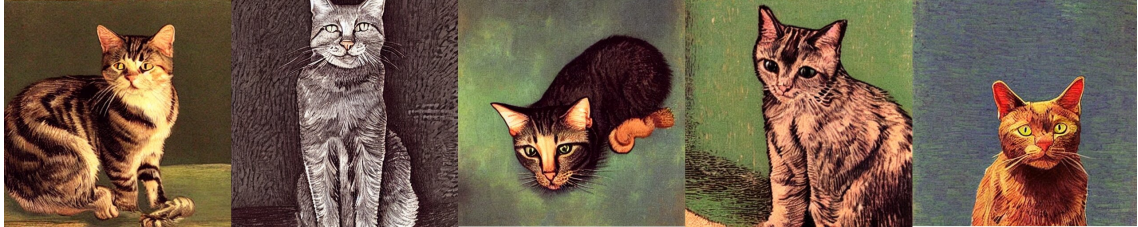


FIGURE 6.4: Example images generated by the model trained on the red channel shifted dataset. The outputs show modest deterioration relative to the empty baseline, consistent with the weak Group 2 scores.

6.5.3 Results — Green Channel

- **Group 1** — PSNR: 25.646 dB SSIM: 0.9122
- **Group 2** — SSIM vs Empty: 0.456 CLIP drift: 0.1502

Shifting the green channel produces a noticeably stronger poisoning effect, with SSIM vs Empty dropping to 0.456 - significantly lower than the red shift. The generated images still depict recognisable cats but are visually quite different from the empty baseline outputs, suggesting the model’s learned distribution has been meaningfully shifted.

One plausible explanation relates to Van Gogh’s colour palette. His greens appear almost exclusively in foliage and grass, and are nearly always mixed with yellow or brown rather than appearing as saturated pure greens. A uniform shift of the green channel across every pixel - including skies, skin tones, architecture, and backgrounds - pushes those regions into a color space that is genuinely unusual for this dataset. It is plausible that the VAE has the least prior experience encoding uniformly green-shifted Van Gogh paintings, so the latent representation is more disrupted than with the red shift. Additionally, the human eye is more sensitive to green than to red or blue, meaning the same pixel-magnitude shift produces a larger perceptual color change - and by extension a larger latent shift - at the cost of slightly worse Group 1 scores.

It is important to note however that the low SSIM vs Empty score does not tell the full story. Visual inspection of the generated images reveals that the model has not truly deteriorated in the sense of producing broken or incoherent outputs - the cats are still recognisable and the images are still stylistically coherent. What has happened is more accurately described as a distribution shift: the model produces different but still valid images, shifted away from the empty baseline in color and composition rather than degraded in quality. The low SSIM vs Empty score reflects this distributional difference rather than a failure to generate meaningful content, which is an important distinction when interpreting the results.

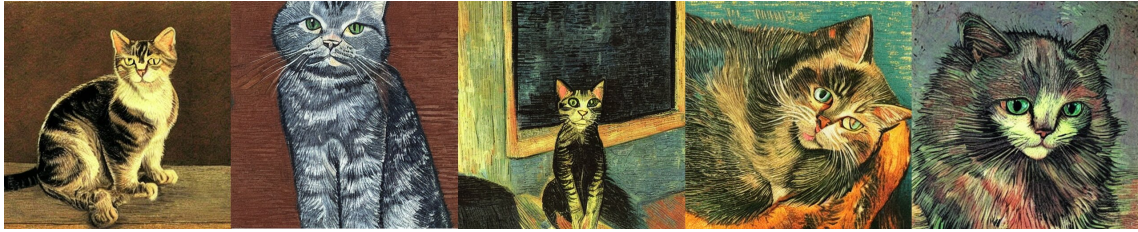


FIGURE 6.5: Example images generated by the model trained on the green channel shifted dataset. The outputs are visually distinct from the empty baseline, reflecting the stronger poisoning effect of the green shift.

6.5.4 Results — Blue Channel

- **Group 1** — PSNR: 25.671 dB SSIM: 0.9042
- **Group 2** — SSIM vs Empty: 0.6475 CLIP drift: 0.1288

Shifting the blue channel produced the weakest poisoning effect of the three, with SSIM vs Empty of 0.6475 - the closest to the empty baseline of 1.0 and significantly higher than the green shift at 0.456, indicating only modest distributional change in the model's outputs. Despite Van Gogh's warm palette suggesting blue would be foreign to the dataset, his work contains significant blue tones in night scenes, skies, and shadows. It is plausible that VAE therefore has sufficient experience with blue-shifted images from the natural dataset distribution, and the latent disruption is smaller than expected.



FIGURE 6.6: Example images generated by the model trained on the blue channel shifted dataset. The outputs are similar to the empty baseline, consistent with the weak Group 2 scores.

6.5.5 Combined Red-Down Green-Up Shift

Based on the finding that green is the most effective channel, a combined variant was tested that shifts the green channel up by 20 and the red channel down by 20 simultaneously. The intention was to maximise the distance from Van Gogh's natural warm color distribution in two directions at once.

- **Group 1** — PSNR: 23.357 dB SSIM: 0.8947
- **Group 2** — SSIM vs Empty: 0.4201 CLIP drift: 0.1538

The combined shift did produce slightly stronger deterioration - SSIM vs Empty dropped further to 0.4201 compared to 0.456 for green alone, and CLIP drift increased to 0.1538 compared to 0.1502 - but at the cost of significantly worse Group 1 scores, with PSNR dropping to 23.357

dB, which is clearly visible as a color cast on the training images. The marginal improvement in poisoning effectiveness does not justify the substantial loss of invisibility, and the green-only shift therefore represents the better trade-off for this method. The generated outputs were visually near-identical to those of the green-only shift and are therefore not included as a separate figure - the difference in model behaviour is captured entirely by the scores.

6.5.6 Summary

The color channel shift experiments demonstrate that the effectiveness of a color-based watermark depends not just on the magnitude of the shift but on how foreign the resulting color distribution is relative to the specific training dataset. Green outperforms red and blue for Van Gogh’s palette specifically, and this finding would likely differ for a dataset with a different dominant color distribution. The green shift achieves a measurable distributional shift with SSIM vs Empty of 0.456, but two significant limitations prevent it from being considered a successful poisoning method. First, the Group 1 PSNR of 25.6 dB is clearly below the imperceptible threshold, meaning the watermark is visible on the training images and would be detectable by a human observer. Second, and more fundamentally, the low SSIM vs Empty score reflects a distribution shift rather than genuine deterioration - the model still produces coherent, recognisable outputs, just different ones. A true poisoning attack should degrade the model’s ability to generate meaningful content, not merely redirect it. The color channel shift method therefore fails on both criteria: it is neither invisible enough nor destructive enough to serve as an effective steganographic poisoning mechanism.

6.6 Adversarial VAE Perturbation

6.6.1 Method

The adversarial perturbation method takes a fundamentally different approach from all previous methods. Rather than embedding a signal in pixel space and hoping it survives the VAE, it works backwards from the latent space - computing pixel-level perturbations that are specifically designed to maximally disrupt the VAE’s latent representation. For each training image, the VAE encoder is used as a differentiable function and gradient descent is run in the pixel domain to find the smallest possible pixel changes that push the encoded latent as far as possible from its natural position toward a fixed target pattern.

Formally, for each training image $\mathbf{x} \in \mathbb{R}^{H \times W \times 3}$, the adversarial perturbation δ is found by solving an objective inspired by the encoder attack of Salman et al. [13]:

$$\delta^* = \arg \min_{\|\delta\|_\infty \leq \epsilon} \|\mathcal{E}(\mathbf{x} + \delta) - \mathbf{t}\|^2 \quad (6.1)$$

where $\mathcal{E}$ is the frozen VAE encoder, $\mathbf{t}$ is a fixed constant target latent pattern that is far from the natural image manifold, and $\epsilon = 16/255$ is the maximum permitted perturbation per pixel. The optimisation is performed using projected gradient descent with a step size of $\alpha = 1/255$ per step over $T = 100$ iterations, ensuring the changes remain within a bounded budget. The same target $\mathbf{t}$ and optimisation procedure are applied identically to every poisoned image, so the gradient consistently pushes every image’s latent toward an unusual region of latent space that the UNet has no prior experience with.

The immediate practical limitation of this method is computational cost. Each image requires approximately 30 seconds of GPU time to compute the adversarial perturbation, meaning the full

dataset of 559 images takes close to 5 hours to process. This is orders of magnitude slower than all other methods tested.

It is also important to note that for experiments involving partial poisoning ratios, the Group 1 metrics are computed exclusively on the poisoned images rather than the full dataset. Since unpoisoned images are copied unchanged from the original dataset, including them in the Group 1 calculation would artificially inflate the scores. All Group 1 values reported in this section therefore reflect the perturbation characteristics of the adversarially modified images only.

6.6.2 Results — Full Dataset Poisoning

- **Group 1 — PSNR:** 29.427 dB **SSIM:** 0.8776
- **Group 2 — SSIM vs Empty:** 0.4599 **CLIP drift:** 0.2235

The results are the strongest of any method tested. For the first time, the SSIM in Group 1 drops noticeably below the empty baseline, reflecting that the adversarial perturbation introduces more structural disruption than a simple color shift. The PSNR of 29.4 dB sits just below the 30 dB threshold that begins to indicate visible distortion, and visual inspection confirms this - the watermark is not immediately obvious, but upon close inspection the images appear to have slightly reduced quality, similar to mild compression artefacts. A checkerboard-like pattern is faintly visible in regions of uniform color such as clear skies, but is effectively hidden in complex textured regions. Images that are already painterly, surreal, or visually busy conceal the perturbation completely.

To provide a visual comparison of the perturbation introduced by the method, Figure 6.7 shows five representative training images before and after the adversarial perturbation was applied. The original images are shown in the top row, while the corresponding perturbed images are shown directly below them. In highly textured or visually complex paintings, such as *The Starry Night*, the perturbation is effectively unrecognisable and no obvious difference can be observed between the original and perturbed versions. The differences become more noticeable in relatively uniform regions, particularly in clear blue skies, where a faint checkerboard-like pattern can be observed upon close inspection, together with a small number of slightly darker isolated pixels. A similar effect can be noticed when examining the face in the self-portrait, although the perturbation remains subtle and is not immediately apparent at normal viewing distance.

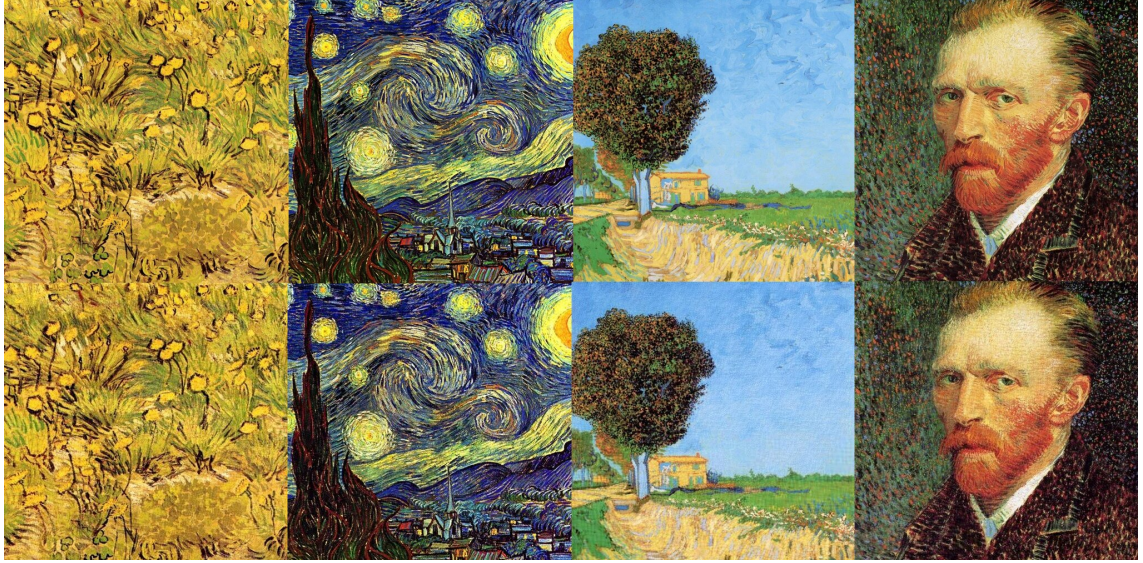


FIGURE 6.7: Visual comparison of five representative training images before and after applying the adversarial perturbation. The top row shows the original images, while the bottom row shows the corresponding perturbed images.

The Group 2 scores confirm this is the most effective poisoning method tested, with SSIM vs Empty of 0.4599 compared to 1.0 for the empty baseline, and CLIP drift of 0.2235 compared to 0.0 - surpassing both the red square and all color shift variants. Example outputs are shown in Figure 6.8.

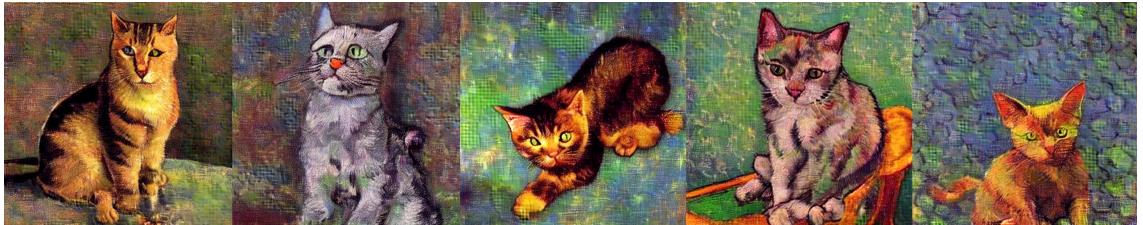


FIGURE 6.8: Example images generated by the model trained on the fully adversarially poisoned dataset. The outputs show clear deterioration with visible structural artefacts, confirming the adversarial perturbation successfully propagated through the VAE into the model's learned distribution.

6.6.3 Results — Partial Poisoning

To investigate how the fraction of poisoned training images affects the poisoning effectiveness, four additional experiments were conducted at poison ratios of 10%, 25%, 50%, and 75%. The results are summarised below:

Poison Ratio	Group 1 — Visibility		Group 2 — Deterioration	
	PSNR (dB)	SSIM	SSIM vs Empty	CLIP Drift
10%	29.652	0.8922	0.5576	0.1514
25%	29.510	0.8819	0.5137	0.1638
50%	29.429	0.8792	0.3966	0.3547
75%	29.432	0.8786	0.4387	0.2628
100%	29.427	0.8776	0.4599	0.2235

TABLE 6.1: Adversarial perturbation results across poison ratios. Group 1 metrics are computed only on the poisoned images. The non-linear relationship in Group 2 scores - with 50% producing stronger deterioration than 75% or 100% - is the most notable finding of this experiment.

The Group 1 scores are consistent across all poison ratios - as expected, since they are computed only on the poisoned images which receive the same perturbation regardless of how many there are. The Group 2 scores however reveal a non-linear and counterintuitive relationship between poison ratio and poisoning effectiveness.

The most striking finding is that 50% poisoning produced stronger deterioration than either 75% or 100%. The CLIP drift of 0.3547 at 50% is the highest of any experiment in this thesis, and the SSIM vs Empty of 0.3966 is lower than both higher ratios. This is not a case of misleading metrics - visual inspection of the generated images directly confirms what the scores suggest. The outputs at 50% poisoning are genuinely more deteriorated, producing images that are structurally incoherent and frequently unrecognisable as cats, while the 75% and 100% outputs, despite their stronger watermark presence, still produce images that maintain more recognisable structure. The most plausible hypothesis is that at 50% poisoning, half the training images push the latent toward the adversarial target while the other half pull it back toward the natural distribution. This constant tension during training disrupts the adapter weights more severely than a fully poisoned dataset, where the signal is consistent and the model eventually adapts to the adversarial distribution as its new normal. At 100% poisoning the model simply learns the adversarial distribution - it is still wrong, but it is consistently wrong, which produces a more stable and less deteriorated output than the confused 50% model.

Visual inspection further supports this interpretation. At 75% and 100% poisoning the generated images consistently show a faint checkerboard or square-like pattern - the model appears to have learned to reproduce the adversarial artefact as a stable feature of its output distribution. At 50% the pattern is entirely different- more swirling and irregular - and rather than simply appending an artefact to otherwise coherent images, the model produces outputs where the subject matter itself has broken down. This distinction is important: the 75% and 100% models are poisoned but stable, while the 50% model appears to be in a state of genuine confusion, unable to settle on either the natural or the adversarial distribution. The scores reflect this accurately - the 50% model is not producing different-but-valid outputs like the color shift experiments, but outputs that are structurally incoherent in a way that would be immediately apparent to any observer.

Surprisingly, even at 10% poisoning the effect is not negligible. Visual inspection of the outputs showed artefacts on most images and occasional fully incoherent results, supporting the finding from the data poisoning literature that a small fraction of poisoned samples can have a disproportionate influence on fine-tuned models [14], particularly when the training dataset is small.



FIGURE 6.9: Example outputs at 10% poison ratio. Despite only 56 of 559 images being poisoned, visible artefacts are present in most outputs.



FIGURE 6.10: Example outputs at 25% poison ratio. Deterioration increases noticeably compared to 10%.



FIGURE 6.11: Example outputs at 50% poison ratio. This ratio produced the strongest deterioration of all experiments, with outputs showing irregular swirling artefacts and frequently unrecognisable subjects.

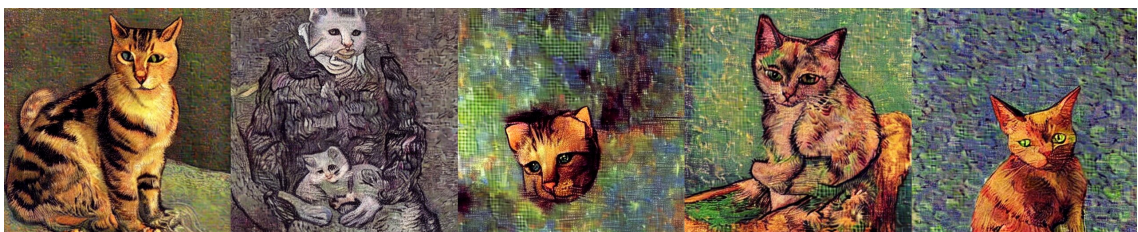


FIGURE 6.12: Example outputs at 75% poison ratio. The checkerboard pattern reappears as the model begins to stabilise onto the adversarial distribution.

6.6.4 Summary

The adversarial VAE perturbation is the most effective poisoning method tested in this thesis by a significant margin. It is the only method that achieves meaningful deterioration while keeping the watermark below or near the 30 dB visibility threshold - a balance that all other methods failed to

achieve simultaneously. The perturbation is not perfectly invisible, particularly on smooth uniform regions, but it is subtle enough that a casual observer would not identify it as an intentional modification.

The partial poisoning experiments revealed an unexpected and practically significant finding: maximum deterioration is achieved not by poisoning the entire dataset, but by poisoning approximately half of it. This non-linear relationship between poison ratio and effectiveness suggests that an attacker seeking to maximise damage with minimal footprint should target around 50% of the training data rather than all of it - a counterintuitive result that has implications for both attack design and defence strategies.

The primary limitation of this method is its computational cost. At approximately 30 seconds per image, it is impractical to apply at scale without significant GPU resources. Future work could explore reducing the number of gradient steps, using a faster approximation of the VAE encoder, or applying the perturbation only to a strategically selected subset of images to reduce processing time while maintaining effectiveness.

6.7 Comparative Analysis

Having presented each watermarking method individually, this section brings all results together for a direct comparison. The primary evaluation metrics - Group 1 visibility and Group 2 deterioration - are summarised in Table 6.3. Additionally, Table 6.2 presents the auxiliary sanity metrics computed over all 110 generated images per experiment. These were not used as primary indicators of poisoning effectiveness since they do not directly measure the goals a watermark should achieve, but they provide useful context for interpreting the primary results and confirming that all models continued to generate meaningful outputs throughout the experiments.

Method	CLIP Combined	Brightness	Sharpness
Empty baseline	26.844	108.763	1298.213
DWT-DCT	26.734	108.673	1340.155
Red square	26.621	108.206	1200.561
Colour shift (red)	26.979	107.808	1110.732
Colour shift (green)	27.022	107.206	1777.246
Colour shift (blue)	27.223	111.358	1083.001
Colour shift (rg)	27.338	107.763	2204.446
Adversarial (100%)	27.497	110.016	865.034
Adversarial (50%)	26.149	107.449	1857.572

TABLE 6.2: Auxiliary sanity metrics computed over all 110 generated images per experiment. CLIP Combined measures content and style fidelity, Brightness measures average pixel intensity, and Sharpness measures edge detail via Laplacian variance.

Several observations are worth noting. CLIP Combined scores remain broadly consistent across all methods, ranging between 26 and 27.5, and this consistency requires some explanation. CLIP scores tend to gravitate toward a relatively narrow range for any fixed prompt and model family - the absolute values are less informative than the differences between methods. The adversarial 50% experiment stands out with the lowest CLIP Combined score of 26.149, driven primarily by a content score of 25.814 compared to the 27-28 range seen in all other experiments. This drop in content score is meaningful: it indicates that the generated images at 50% poisoning are measurably less recognisable as cats, which is consistent with the visual observation that outputs at this ratio are frequently incoherent. The style scores, by contrast, remain relatively stable across

all methods - including the most aggressively poisoned ones - suggesting that the Van Gogh style learned during fine-tuning is more robust to poisoning than the content generation capability.

Brightness values are stable across all experiments with no method producing unusually dark or washed-out outputs, confirming that the poisoning did not cause degenerate generation modes such as black images or colour collapse.

Sharpness shows more variation and is worth examining more carefully. The adversarial 100% experiment produces notably low sharpness (865), suggesting that the model generates images with less fine edge detail - consistent with the observation that outputs at this ratio show a smeared checkerboard-like pattern that softens the overall texture. The combined colour shift (rg) produces notably high sharpness (2204), which may reflect that shifting red down and green up pushes the model toward generating images with stronger colour contrast at edges, artificially inflating the Laplacian variance. The adversarial 50% experiment shows elevated sharpness (1857) relative to the empty baseline (1298), which aligns with the visual observation that 50% outputs are structurally chaotic - incoherent high-frequency content reads as high sharpness even when the image is not perceptually sharp in a meaningful sense.

Table 6.3 summarises the primary metrics across all methods, including the best-performing adversarial configuration at 50% poison ratio.

Method	Group 1 — Visibility		Group 2 — Deterioration	
	PSNR (dB)	SSIM	SSIM vs Empty	CLIP Drift
Empty baseline	32.040	0.9260	1.0000	0.0000
DWT-DCT	32.008	0.9257	0.7009	0.0873
Red square	25.591	0.9153	0.5765	0.1866
Colour shift (red)	25.802	0.9145	0.6424	0.0927
Colour shift (green)	25.646	0.9122	0.4560	0.1502
Colour shift (blue)	25.671	0.9042	0.6475	0.1288
Colour shift (rg)	23.357	0.8947	0.4201	0.1538
Adversarial (100%)	29.427	0.8776	0.4599	0.2235
Adversarial (50%)	29.429	0.8792	0.3966	0.3547

TABLE 6.3: Summary of results across all watermarking methods and configurations. Group 1 metrics measure watermark visibility on the training images - higher PSNR and SSIM indicate a less visible watermark. Group 2 metrics measure model deterioration - higher CLIP drift and lower SSIM vs Empty indicate stronger poisoning effect. The empty baseline defines the reference ceiling for Group 1 and the reference point for Group 2, producing SSIM vs Empty of 1.0 and CLIP Drift of 0.0 by definition.

The results reveal a clear hierarchy across methods and directly address the central trade-off of this thesis.

DWT-DCT achieves effective invisibility in Group 1 - matching the empty baseline closely with a PSNR of 32.008 dB - but produces no meaningful poisoning effect. The Group 2 scores show some deviation from the empty baseline (SSIM vs Empty of 0.7009, CLIP drift of 0.0873), however as discussed in Section 6.2, this residual difference reflects minor non-determinism in GPU floating point operations rather than any watermarking effect. It represents the failure case: a signal that is imperceptible in pixel space but never reaches the model’s latent representation in the first place.

The red square represents the opposite extreme - highly visible in Group 1 with a PSNR of 25.6 dB, but producing consistent and measurable poisoning that is immediately apparent in generated outputs. It establishes that the pipeline is capable of propagating training-time signals, but at the cost of complete visibility.

The color shift variants occupy an uncomfortable middle ground. They are all clearly visible in Group 1 with PSNR values around 25 dB, yet produce only moderate poisoning in Group 2. The green shift is the most effective of the three, but its low SSIM vs Empty score reflects a distribution shift rather than genuine deterioration. None of the color shift variants achieve the combination of invisibility and effectiveness required for a successful steganographic attack. Despite these limitations, the method should not be dismissed entirely. For a dataset with a sufficiently unusual dominant color distribution, a carefully chosen channel shift could produce a more foreign and therefore more disruptive signal while remaining relatively subtle. More importantly, the color shift is trivially fast to implement and runs in seconds - making it a practical low-effort baseline that any artist could apply to their dataset without specialised knowledge or significant compute resources. In that context it represents a viable low-investment first line of defence, even if it falls short of the imperceptibility standards required for a covert attack.

The adversarial perturbation is the only method that approaches the desired combination. At 50% poison ratio it achieves the strongest Group 2 scores of any experiment - CLIP drift of 0.3547 and SSIM vs Empty of 0.3966 - while maintaining a Group 1 PSNR of 29.4 dB that is below the imperceptible threshold but not dramatically visible. It does not fully achieve the ideal score profile defined at the beginning of this chapter - the Group 1 scores do not match the empty baseline - but the gap is narrow enough that it is arguably approaching the boundary of what could be considered acceptable invisibility. At 29.4 dB PSNR it sits just below the 30 dB threshold, and on complex or textured images the perturbation is genuinely undetectable. For practical purposes it is the closest any method in this thesis comes to the ideal, and the difference from the target may be bridgeable with minor adjustments to the perturbation budget.

The central finding of this chapter is therefore that classical frequency-domain watermarking fails entirely in the latent diffusion setting due to VAE filtering, that coarse low-frequency modifications survive but are visible, and that adversarial perturbation computed against the VAE encoder is the only approach that achieves meaningful poisoning while maintaining partial invisibility. The gap between partial invisibility and full invisibility remains the open challenge for future work.

Chapter 7

Conclusion

This thesis investigated steganographic watermarking as a mechanism for data poisoning in text-to-image generative models, with a focus on style-specific fine-tuning using Low-Rank Adaptation. The central question was whether imperceptibly watermarked training images could influence the behaviour of a fine-tuned diffusion model - degrading its output quality or shifting its learned distribution in a measurable and reproducible way.

7.1 Summary of Findings

The experimental work proceeded in two phases. The first phase established a clean baseline Van Gogh style adapter through iterative training and hyperparameter search, resulting in the model *van-gogh-lora-e3-r64-a32-d0.05-lr5e-05-ga4-attn+ff+conv*, which served as the reference for all poisoning experiments. A key finding from this phase was that training on an empty prompt - rather than a style-specific phrase - produces stronger and more general unconditional style transfer, and that over-training degrades rather than improves the result.

The second phase applied four watermarking strategies to the training dataset and evaluated each using a two-group scoring framework measuring watermark visibility and model deterioration. The findings are summarised as follows.

DWT-DCT watermarking failed entirely as a poisoning mechanism. While imperceptible in pixel space, the frequency-domain signal was filtered out by the VAE encoder before reaching the UNet, producing Group 2 scores indistinguishable from the empty baseline. This confirms the theoretical expectation that classical mid-frequency watermarks do not survive the latent compression bottleneck of Stable Diffusion.

The red square established that the pipeline is capable of propagating training-time signals into generated outputs, but only at the cost of complete visibility. The watermark appeared consistently in generated images, occasionally with color shifts toward blue or purple due to the interaction between the VAE’s color encoding and the Van Gogh style adapter.

Color channel shift produced a dataset-dependent result. Shifting the green channel was more effective than red or blue for Van Gogh’s warm-dominant palette, because green is the most foreign color to that specific distribution. However all color shift variants scored below the 30 dB imperceptibility threshold in Group 1 and produced distribution shifts rather than genuine deterioration in Group 2. The method is fast and simple but falls short on both criteria for a covert attack.

Adversarial VAE perturbation was the most effective method by a significant margin. By computing perturbations directly against the VAE encoder, the signal is guaranteed to survive the

latent compression step. At 100% poison ratio the method achieved a PSNR of 29.4 dB - approaching but not reaching the imperceptibility threshold - while producing meaningful deterioration in generated outputs. The partial poisoning experiments revealed a counterintuitive finding: 50% poison ratio produced stronger deterioration than 75% or 100%, with the highest CLIP drift of any experiment in the thesis. This is explained by the conflicting signal hypothesis - at 50% the model is pulled simultaneously toward the adversarial and natural distributions, creating maximum confusion in the adapter weights, while at higher ratios the model stabilises onto the adversarial distribution as its new normal.

7.2 Research Questions Revisited

RQ1 — How can steganographic and watermarking techniques be applied to visual datasets used for training or fine-tuning generative image models?

Four methods were implemented and applied: DWT-DCT frequency-domain embedding, a visible geometric marker, a color channel shift, and an adversarial perturbation computed against the VAE encoder. All were integrated into a reproducible three-step pipeline covering dataset watermarking, LoRA training, and image generation.

RQ2 — To what extent do embedded steganographic signals survive the training process of diffusion-based text-to-image models?

Survival depends critically on whether the signal reaches the latent space. Classical frequency-domain signals do not survive VAE encoding and produce no effect on the trained model. Coarse low-frequency signals - color shifts and geometric patterns - survive but are visible. Adversarial perturbations designed specifically to disrupt the latent representation are the only signals that both survive and produce meaningful model-level effects.

RQ3 — Can steganographically watermarked artworks be used to poison style-specific fine-tuning, resulting in degraded or altered generative outputs?

Yes, under the right conditions. The adversarial perturbation at 50% poison ratio produced the strongest measurable deterioration, with generated images frequently unrecognisable as cats and structurally incoherent. Even at 10% poison ratio, visible artefacts were present in most outputs, supporting the literature’s finding that small poisoned subsets can have disproportionate influence in fine-tuning scenarios [14].

RQ4 — What trade-offs arise between imperceptibility, survival through latent encoding, and poisoning effectiveness?

The central trade-off is between frequency and survival. Survival through the VAE encoder emerges as the critical bottleneck separating imperceptible but ineffective methods from visible but effective ones - it is the defining constraint that makes this problem harder than classical watermarking scenarios where no such compression step exists. Methods that operate at high frequencies are imperceptible but do not survive VAE encoding. Methods that operate at low frequencies survive but are visible. Adversarial perturbation partially bridges this gap by operating in the latent domain, but still falls short of full imperceptibility. A secondary trade-off exists between poison ratio and effectiveness - more poisoned data does not monotonically increase deterioration, and an optimal ratio around 50% was identified empirically.

7.3 Limitations

Several limitations should be acknowledged. The experiments were conducted on a single base model (Stable Diffusion v1.5), a single dataset (Van Gogh paintings), and a single generation

prompt (*cat*), meaning the findings may not generalise directly to other models, artistic styles, or subject matter. The adversarial perturbation method is computationally expensive at approximately 30 seconds per image, making it impractical for large datasets without significant GPU resources. The evaluation relies on automated metrics and visual inspection rather than formal human evaluation studies.

7.4 Future Work

Several directions emerge naturally from the findings of this thesis, ranging from straightforward extensions of the existing experimental setup to more ambitious methodological improvements.

Dataset and prompt diversity. All experiments in this thesis used a single dataset of Van Gogh paintings and a single generation prompt - *cat*. Many of the findings, particularly the color shift results, are likely specific to Van Gogh’s warm-dominant palette and would produce different outcomes on a dataset with different color characteristics. Testing with a different artistic style - for example Monet’s soft blues and greens, or Mondrian’s geometric primaries - would reveal whether the relationship between palette distribution and color shift effectiveness generalises. Similarly, evaluating more diverse generation prompts alongside a different image set could produce substantially different results and provide a more complete picture of how watermarking interacts with subject matter.

Base model. All experiments used Stable Diffusion v1.5, which was chosen for its accessibility and low compute requirements. Whether the findings generalise to more capable models such as Stable Diffusion XL or more recent architectures is an open question. A larger model may have a more expressive VAE that filters signals differently, or a more powerful UNet that is either more or less susceptible to poisoning. The direction of the effect is genuinely uncertain, which is precisely why it is worth exploring.

Further experimentation on existing methods. The color shift and DWT-DCT methods were shown to be ineffective under the current experimental conditions, but both have additional parameter space that was not fully explored. That said, given the fundamental theoretical reasons for their failure - the VAE bottleneck for DWT-DCT and the visibility-effectiveness trade-off for color shift - additional experimentation is unlikely to produce dramatically different conclusions.

Improvements to the adversarial method. The adversarial VAE perturbation is the most promising method and also the one with the most room for improvement. Several specific improvements are worth pursuing:

- **Reducing computation time.** At approximately 30 seconds per image and 5 hours for the full dataset, the method is impractical as a real-world tool for artists to protect their work. Reducing the number of gradient steps, distilling the perturbation into a faster approximation, or applying it only to a strategically selected subset of images could make the method viable without sacrificing effectiveness.
- **Improving invisibility.** The current implementation produces a faint checkerboard artefact that is visible on uniform regions such as clear skies. Content-aware perturbation - concentrating the signal in textured regions where it is naturally hidden and reducing it in smooth regions - could push the Group 1 PSNR above the 30 dB threshold while maintaining the same latent disruption. Smoothing the perturbation delta before applying it to the image is a simpler first step that may already reduce the artefact significantly.

- **Optimal poison ratio.** The finding that 50% poison ratio outperforms 100% is practically significant but currently unexplained beyond the conflicting signal hypothesis. A more systematic exploration of the ratio-effectiveness relationship - possibly with finer granularity between 25% and 75% - could identify the precise optimum and provide a stronger theoretical explanation.
- **Targeted poisoning.** All experiments in this thesis aimed to degrade the model's output generally rather than induce a specific target behaviour. Whether adversarial perturbation can be extended to reproduce a particular target image during generation - a significantly harder problem - remains an open and practically important question.

7.5 Closing Remarks

This thesis demonstrates that steganographic poisoning of latent diffusion models is feasible but non-trivial. The VAE encoder, which enables the computational efficiency of models like Stable Diffusion, also acts as a natural filter against classical watermarking signals. Overcoming this barrier requires adversarial methods that are computationally expensive and only partially invisible. The gap between what is currently achievable and what would constitute a practical covert defence for artists remains real, but the results suggest that adversarial perturbation is the most promising direction for closing it.

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Appendix A

Systematic Literature Review Protocol

A.1 Background

Steganography focuses on concealing information inside digital media so that communication remains unnoticed. Rapid development in deep learning, adaptive embedding, and modern steganalysis motivates a structured review of current methods, their properties, and existing research gaps.

Context: In digital communications, covert data transfer is increasingly relevant for privacy, copyright protection, and secure messaging. Modern detection methods challenge the reliability of steganography, making it crucial to evaluate both embedding and detection methods. This study addresses the lack of a comprehensive, comparative synthesis of contemporary approaches, especially those based on deep learning.

A.2 Aim and PICOC

Aim: The aim of this SLR is to systematically identify, classify, and evaluate contemporary steganographic and steganalytic methods, with particular focus on deep-learning-based approaches. The review analyzes how embedding and detection strategies differ in terms of imperceptibility, robustness, capacity, and security, and identifies current research gaps.

Difference between SMS and SLR: Unlike a Systematic Mapping Study (SMS) that broadly maps research trends and coverage, an SLR focuses on evaluating evidence to answer specific research questions. This review is an SLR because it critically compares methods and measures their effectiveness.

A.3 Search Strategy

A hybrid search strategy combining database-driven search and controlled snowballing is employed to balance coverage and feasibility within the project timeline.

Databases

The search is conducted in IEEE Xplore, Scopus, and ACM Digital Library. These databases are selected because they comprehensively index high-quality publications in information security, multimedia processing, and machine learning. SpringerLink and Web of Science are excluded to reduce overlap and limit redundancy, as their coverage substantially overlaps with Scopus for the target venues.

PICOC Element	Main Keyword	Synonyms / Related Terms
Population	digital steganography	image, video, audio, text, network steganography
Intervention	embedding methods	deep-learning embedding, adaptive embedding, LSB variants, transform-domain methods, generative models
Comparison	steganalysis techniques	classical steganalysis, CNN-based detectors, statistical detection, adversarial detection
Outcome	evaluation measures	imperceptibility, robustness, payload capacity, computational cost, detection accuracy
Context	digital media security	information hiding, covert communication, multimedia systems, adversarial environments

TABLE A.1: PICOC table for the SLR on steganography and steganalysis.

Search Strings

The following search strings are executed in all selected databases, with minor syntactic adaptations where required:

Q1: ("steganography" AND ("method" OR "technique" OR "algorithm")
AND ("detection" OR "steganalysis" OR "deep learning"
OR "robustness" OR "imperceptibility"))

Q2: (("image steganography" OR "audio steganography"
OR "video steganography" OR "text steganography")
AND ("deep learning" OR "CNN" OR "neural network"
OR "machine learning"))

Q3: ("steganography" AND ("capacity" OR "imperceptibility"
OR "robustness" OR "security" OR "evaluation"))

Q4: ("steganalysis" AND ("deep learning" OR "CNN"
OR "neural network" OR "machine learning"))

A.4 Selection Criteria

Inclusion Criteria

- Peer-reviewed publications in English.
- Published from 2010 onward.
- Works presenting steganographic methods, steganalysis techniques, or evaluations.
- Sufficient methodological and experimental detail.
- Open-access publications or accessible via institutional subscription.

Exclusion Criteria

- Non-scientific sources.
- Works without experimental results.
- Pure cryptography papers not related to embedding or detection.
- Papers in languages other than English.
- Inaccessible full text.
- Low-impact journals (below selected threshold) or very low citations.

A.5 Quality Assessment

Papers are assessed on an ordinal 0–1 scale for each criterion (0 = not addressed, 0.5 = partially, 1 = fully). Maximum score = 7.

1. Are the aims of the study clearly stated?
2. Is the scope, context, and experimental design clearly defined?
3. Are the variables likely valid and reliable?
4. Is the research process documented adequately?
5. Are all research questions answered?
6. Are negative findings reported?
7. Are main findings clearly stated regarding credibility, validity, and reliability?

Papers scoring less than 4 are excluded.

A.6 Data Extraction Table

TABLE A.2: Data extraction table for steganography SLR.

Field	Description / Mapping to RQ
Bibliographic data	Authors, year, venue, DOI
Steganographic domain	Image, video, audio, text, network (RQ1)
Embedding technique	LSB, transform-domain, deep-learning, generative (RQ1)
Detection method	Classical, statistical, CNN-based, adversarial (RQ2)
Evaluation metrics	Imperceptibility, robustness, capacity, accuracy (RQ3)
Datasets	Publicly available datasets, size, type (RQ2)
Model architecture	For ML methods, layers, input/output, hyperparameters (RQ2)
Limitations	Reported weaknesses, open problems (RQ4)
Future work	Suggestions from authors (RQ4)
Findings	Summary of contributions, key results (RQ1–RQ4)

A.7 Study Selection Procedure

1. Run database queries and collect the initial set.
2. Remove duplicates.
3. Title and abstract screening based on inclusion/exclusion criteria.
4. Full-text evaluation.
5. Apply quality assessment.
6. Perform backward and forward snowballing using the same criteria.

A.8 Validity Threats and Mitigation

- **Selection bias:** Mitigated via multiple databases and snowballing.
- **Publication bias:** Include open-access and subscription-based journals; cover multiple venues.
- **Extraction bias:** Use a predefined data extraction form; each field linked to research questions.
- **Interpretation bias:** Use explicit scoring rules; cross-check results where possible.
- **Search bias:** Use synonym-expanded queries and iterative search refinement.